STAFF REVIEW · H446 039 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. 6

2. The oldest item

3. To avoid non-termination

4. 1

I Do — source worked steps

1. Initialise and trace pass one — Initially total=0 and i=1. The test 1<=3 is true; total becomes 1, then i becomes 2.

2. Trace the remaining passes — At i=2, total becomes 1+2=3; i becomes 3. At i=3, total becomes 3+3=6; i becomes 4.

3. Show the failed test — 4<=3 is false, so the body stops and total remains 6. A while loop may run zero times; a do...while tests after its body and runs at least once.

We Do — guided CHECK

1. Check your thinking
3: i=0,1,2.

2. Discuss, then respond
while (x < 5) x++; starting x=3: final x?
Check your thinking
5.

You Do — source mark schemes

1. 0 → 2 → 6 → 12.

2. When repetitions depend on a condition rather than a fixed count.

3. for (int i=1; i<=3; i++) Console.WriteLine(i);

Exit Ticket — source answer

1. False. A while loop may execute zero times if its condition is false at the start.